MLOps Assignment 2

Tracking a Pre-trained Sentiment Model with MLflow

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Objective

To track a pre-trained sentiment-analysis model using MLflow, record experiment parameters and performance metrics, save prediction outputs as an artifact, and compare runs using different batch sizes.

Tools and Technologies

Python 3.11.9, MLflow 3.17.2, Hugging Face Transformers, PyTorch, Pandas, and a pre-trained DistilBERT sentiment-analysis model.

Experiment Summary

The same model and 10 labelled sentences were evaluated in three runs with batch sizes 1, 8, and 4. The experiment logged accuracy, average confidence, total inference latency, parameters, and the predictions.csv artifact in MLflow.

Recorded Results

Run name	Batch size	Accuracy	Avg. confidence	Latency (s)
run_batch1	1	100.00%	0.9997	0.272
run_batch8	8	100.00%	0.9997	0.122
run_batch4	4	100.00%	0.9997	0.138

Short Report

1. Which run had the lowest latency, and why might that be?

run_batch8 had the lowest measured latency at 0.122 seconds, followed by run_batch4 at 0.138 seconds and run_batch1 at 0.272 seconds. Processing several sentences in a batch can reduce repeated inference overhead. Exact timings can vary with model warm-up and system conditions.

2. Did accuracy change between runs? Why or why not?

No. Accuracy remained 1.00 (100%) for all three batch sizes. The same pre-trained sentiment-analysis model and the same 10 test sentences were used. Batch size changes how inputs are grouped for inference, but it does not change the model weights.

3. Name two benefits of using MLflow instead of printing results in the terminal.

1. Reproducibility and record-keeping: MLflow stores parameters, metrics, and artifacts such as predictions.csv so they can be inspected later.
2. Easy comparison: MLflow displays runs and their metrics side by side, making it easier to compare accuracy, confidence, and latency than reading terminal output alone.

Program Execution

The terminal output below records the successful completion of all three experiment runs and their measured metrics.

```
PS C:\Users\Syeda Afira Ayman> cd MLOps_Assignment_2
PS C:\Users\Syeda Afira Ayman\MLOps_Assignment_2> .\mlops-env\Scripts\Activate.ps1
(mlops-env) PS C:\Users\Syeda Afira Ayman\MLOps_Assignment_2> notepad mlflow_lab.py
(mlops-env) PS C:\Users\Syeda Afira Ayman\MLOps_Assignment_2> python mlflow_lab.py
2026/10/29 17:52:19 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experi
ment.

Starting run_batch1 with batch size 1
Loading weights: 100%|| 104/104 [00:00<00:00, 489.17it/s]
Run: run_batch1
Accuracy: 100.00%
Average confidence: 0.9997
Total latency: 0.272 seconds

Starting run_batch8 with batch size 8
Loading weights: 100%|| 104/104 [00:00<00:00, 500.56it/s]
Run: run_batch8
Accuracy: 100.00%
Average confidence: 0.9997
Total latency: 0.122 seconds

Starting run_batch4 with batch size 4
Loading weights: 100%|| 104/104 [00:00<00:00, 436.12it/s]
Run: run_batch4
Accuracy: 100.00%
Average confidence: 0.9997
Total latency: 0.138 seconds

All three experiments completed!
```

Figure 1. Python execution output showing run_batch1, run_batch8, and run_batch4.

MLflow Runs and Metric Charts

The MLflow runs page displays the three recorded runs and visual comparisons for accuracy, average confidence, and total latency.

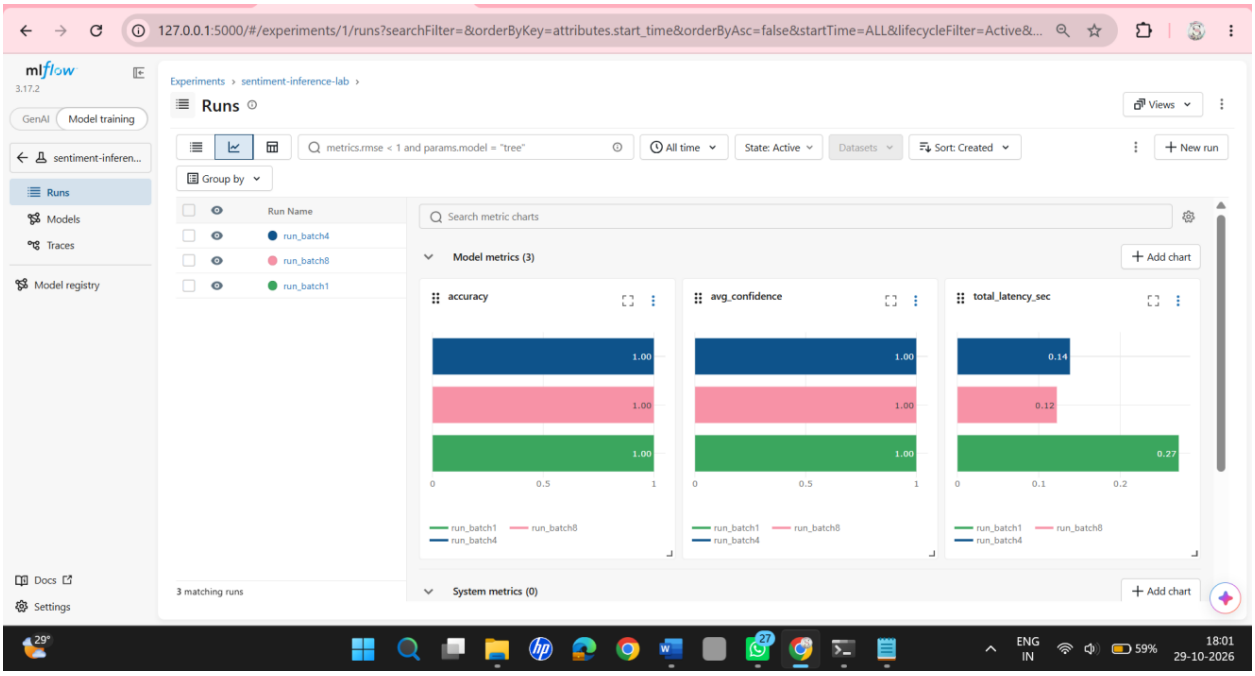


Figure 2. MLflow runs page with metric charts for the three batch-size experiments.

Run Comparison and Artifact

The comparison view confirms the logged parameters and metrics for each run. The artifact section lists predictions.csv.

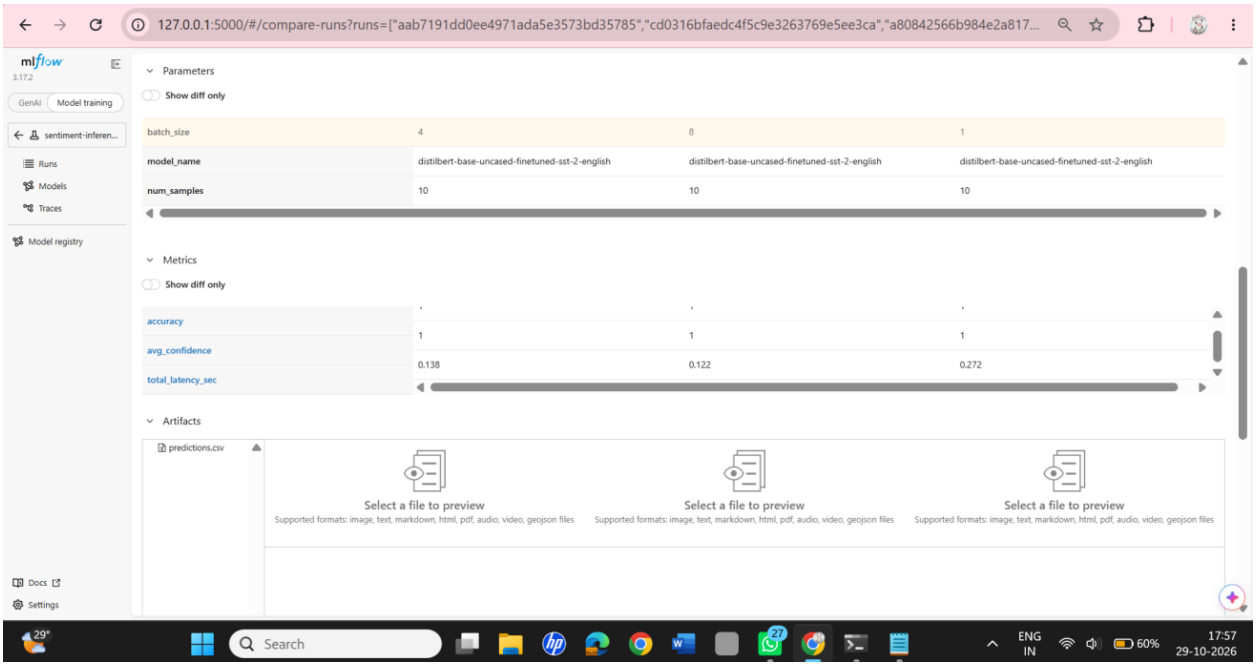


Figure 3. MLflow comparison details showing parameters, metrics, and predictions.csv.

Comparison Visualization

The comparison visualization provides a side-by-side view of batch size and accuracy for the selected runs.

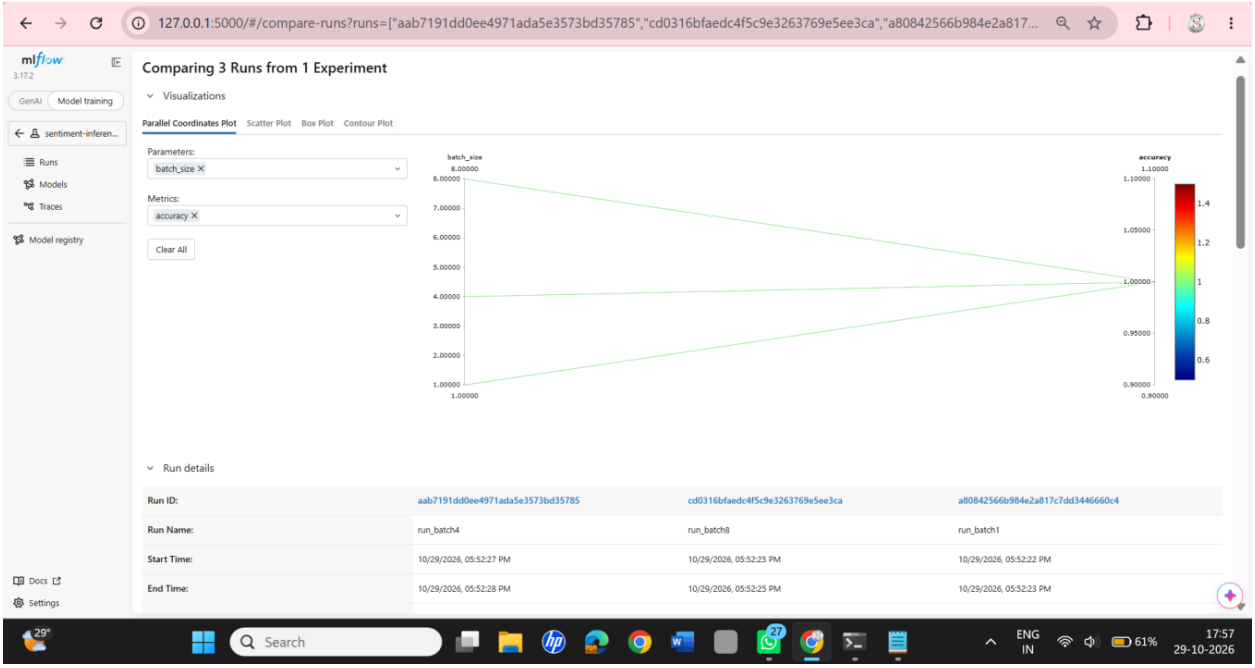


Figure 4. MLflow comparison visualization for the three experiment runs.